

TARGET COMPANY: NESTLÉ S.A.

DAY 09: OEE & 6 BIG LOSSES ANALYSIS REPORT

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Project Scope: OEE Audit & Quality Control Analysis

Project Context: Part of 100-Day Self-Learning Challenge Series (Personal Knowledge & Skill Growth)

1. TARGET INDUSTRY, PROBLEM STATEMENT & AUDIT OBJECTIVE

Target Company: Nestlé S.A.

Target Industry: Fast-Moving Consumer Goods (FMCG) / Food & Beverage Packaging

Target Production Line: Automated VFFS High-Speed Eco-Friendly Packaging Line

During the transition of a high-speed Vertical Form-Fill-Seal (VFFS) snack packaging line from standard flexible plastic to eco-friendly biodegradable pouch material, key operational challenges were identified during shift execution.

The Core Operational Challenges Identified:

- **Sealing Leakage & Efficiency Losses:** Thermal sensitivity of biodegradable film causes seam line weakness upon minor temperature fluctuations, resulting in continuous pouch sealing leaks and line efficiency drop.
- **Environmental & Regulatory Compliance:** Non-biodegradable plastic usage poses environmental risks; fixing pouch integrity issues is mandatory to comply with **ISO 14001** (Environmental Management System) standards.

Main Objectives of the OEE Audit:

1. Reduce unplanned machine downtime by replacing breakdown maintenance with proactive checks.
2. Eliminate product scrap and rework arising from defective heat sealing.
3. Optimize line throughput and overall shopfloor productivity.
4. Ensure strict adherence to ISO 14001 environmental sustainability guidelines.

2. SHIFT OPERATIONAL DATA & STEP-BY-STEP OEE CALCULATIONS

Operational numbers recorded from an 8-hour shift execution on the line:

Given Input Data (Collected from Shift Logs):

- **Total Shift Duration:** 8 Hours = 480 Minutes
- **Planned Breaks (Lunch/Tea):** 30 Minutes
- **Planned Production Time:** 480 - 30 = **450 Minutes**
- **Unplanned Downtime:** 60 Minutes (approx 7.5 mins lost per hour due to sealing bar adjustments and film alignment)
- **Actual Operating Time:** 450 - 60 = **390 Minutes**
- **Target Machine Speed:** 100 Pouches per Minute (PPM)
- **Expected Output in 390 Mins:** 390 × 100 = **39,000 Pouches**
- **Total Output Produced:** **39,000 Pouches**
- **Rejection Rate:** 5% (due to improper seal leakage and weight variation)
- **Rejected Pouches (Scrap):** 39,000 × 0.05 = **1,950 Pouches**
- **Good Pouches (First Time Right):** 39,000 - 1,950 = **37,050 Pouches**

Step-by-Step OEE Metric Calculations:

1. Availability Rate (%)

Formula: (Operating Time / Planned Production Time) × 100

Calculation: (390 / 450) × 100 = **86.67%**

2. Performance Rate (%)

Formula: (Actual Output / Target Output) × 100

Calculation: (39,000 / 39,000) × 100 = **100.00%**

3. Quality Rate (%)

Formula: (Good Pouches / Total Output) × 100

Calculation: (37,050 / 39,000) × 100 = **95.00%**

Final Overall OEE Score:

OEE = Availability × Performance × Quality

OEE = 0.8667 × 1.0000 × 0.9500 × 100 = **82.33%**

Audit Inference: World-class target benchmark is 85%. Current performance at 82.33% highlights the need to address downtime losses and sealing defects.

3. ACTION PLAN & IMPROVEMENT STRATEGY FOR 6 BIG LOSSES

To eliminate operational losses and elevate OEE above 85%, a structured 3-tier action plan was designed:

Pillar Category	Tool / Advanced Technology Implemented	Key Operational Benefit & Impact
AI Smart Tech	AI Material & ISO Compliance Sensor	Scans film thickness and verifies ISO eco-sustainability certified material before feeding.
Predictive AI	Real-time AI Loss Analytics Sensor	Detects efficiency drop immediately and pops up instant corrective advice for operator screen.
Training	Workforce Operator Training Program	Teaches operators how to handle new AI sensors and controls to get maximum output without errors.
RCA Tools	Fishbone (Ishikawa) & 5-Why Analysis	Finds real root cause of frequent sealing bar stops and temperature fluctuations.
Risk Prevention	Process FMEA (PFMEA)	Identifies potential sealing failure modes in advance to reduce RPN score.
Kaizen	PDCA Cycle & SMED Integration	Reduces film roll changeover downtime and supports daily continuous shopfloor improvement.

4. STANDARDIZED DOCUMENTATION & ISO GOVERNANCE FRAMEWORK

- To ensure systemic governance and audit readiness, three core documentation frameworks were structured:
- 1. **SOP (Standard Operating Procedures) File:** Standard operating guidelines for packaging equipment, daily thermal bar maintenance, and AI sensor handling.
 - 2. **Daily Audit Report File:** Logbook tracking shift OEE indices, 6 Big Losses breakdowns, scrap quantities, and sensor performance metrics.
 - 3. **ISO Standards Compliance Report:** Mapping shopfloor execution directly to international ISO frameworks:
 - **ISO 9001:** Quality management framework for raw material integrity and First-Time-Right production.
 - **ISO 14001 & ISO 14004:** Operational guidelines for Environmental Management Systems (EMS) and sustainable packaging compliance.
 - **ISO 14011 & ISO 14012:** Governance procedures for internal quality/environmental audits and auditor competencies.
 - **ISO 14020, 14021, 14022 & 14023:** Eco-labeling and self-declared environmental claim validation for biodegradable polymers.

Self-Learning & Knowledge Ownership Declaration:
This OEE report and industrial audit framework has been prepared by **Jagriti Singh** through independent technical research and analytical study. Executed as part of the **100-Day Self-Learning Challenge Series** to strengthen core competencies in Quality Engineering, Total Quality Management (TQM), and Sustainable Manufacturing.